Does and how does Income Inequality Influence Violent Crime Rates in Baltimore from 2012 to 2017 in a significant way?

Charlotte Jin

University of Toronto

April 5, 2025

Abstract

What factors contribute to the violent crime rate in urban neighbourhoods? This paper explores new diterminants on the economic and social factors of violent crime in Baltimore from 2012 to 2017, using community-level data from OpenBaltimore and over 2,000 mental health service information scraped from the 988 Suicide and Crisis Lifeline. I used Ordinary Least Squares (OLS) and machine learning models to capture linear and non-linear relationships. While OLS demonstrates a negative correlation between income and violent crime, I found that socio-environmental factors, especially liquor export density and juvenile arrest rates, were stronger predictors. Regression trees and random forests indicate significant interaction effects and thresholds not detected in traditional models. Notably, low-income neighbourhoods with high levels of liquor outlets and juvenile-related stressors face disproportionately high crime rates. These findings suggest that economic hardship and social disorder shape crime. Effective policy must target both dimensions to reduce the root cause of violent crime in high-risk neighbourhoods.

JEL Codes: C55, I38, K42, R23

1. Introduction

With violent crime being a growing concern in modern society, understating the correlation between income levels and the frequency of violent crime in economic conditions is significant for policymaking (Fajnzylber et al., 2002). This study examined whether and how income inequality significantly impacted the violent crime rates in Baltimore from 2012 to 2017. Previous research has explored the relationship between economic inequality and crime, highlighting the effect of unemployment, education and income disparities (Sampson et al., 1997). Unlike previous research that has established a general link between economic factors and crime at the national or transnational levels (Yildiz et al., 2013), this study focuses on community-level crime patterns to provide a perspective on a specific city.

To understand the roots of violent crime, imagine a young man, John, living in a low-income neighbourhood in Baltimore. With limited opportunities for education and jobs, John weighed the risks and benefits of crime activity, a choice based on Becker's (1968) criminal economic theory. Grogger (1998) explains that lower wages and job scarcity increase this risk, especially for young people. Gillani, Rehman and Gill (2009) found that juvenile crimes respond sensitively to wage changes, strengthening this relationship. However, economics alone cannot tell the whole story. Chalfin and McCrary (2017) point out that social factors such as policing and neighbourhood stability also determine whether someone like John will turn to crime.

All the datasets used in this research come from the OpenBaltimore website, which involves economic indicators such as household income, unemployment rate, dropout rate, mortgage foreclosure rates, juvenile arrest rate, liquor outlet density, CitiWatch cameras and neighbourhood classification. Income level is the main predictor, as economic hardship is often considered a primary driving factor of criminal activity (Raphael & Winter-Ebmer, 2001). this study introduced the mental health service data from the 988 Suicide and Crisis Lifeline to recognize the complex social background. These data provide a deeper insight into inequality in access and the availability of social infrastructure services. This study also applies machine learning to reveal non-linear relationships and hidden interaction effects in predicting violent crime.

Serval hypotheses emerge from this framework. Firstly, low-income neighbourhoods are expected to experience higher rates of violent crime due to economic regression, unemployment and educational challenges. Secondly, if economic conditions improve over time, violent crime rates may decline, and persistent economic disparities may sustain high crime rates (Ehrlich, 1973). Thirdly, structural factors such as foreclosure rates and liquor export density may simulate crime rates by promoting social disorder (Levitt, 1996). By focusing only on violent crime, this study aims to contribute a more nuanced understanding of crime patterns by identifying the most related economic and social factors, avoiding a general and predictable conclusion.

2. Data and Methods

2.1 Data Selection

The datasets used in this study were collected primarily from the OpenBaltimore government website and covered from 2012 to 2017. The dependent variable is the violent crime rate, including homicide, rape, aggravated assault and robbery, measured by the number of violent crimes per 1000 residents at the neighbourhood level. Independent variables include income level, unemployment rate, high school dropout rate, total population, mortgage foreclosure rates, juvenile arrest rate, liquor outlet density, and CitiWatch camera locations. Income level is the main predictor, as economic differences affect crime patterns. Unemployment and dropout rates were included to assess whether a lack of economic and educational opportunities may lead to higher crime rates, while the total population controlled for differences in community size. Mortgage foreclosure rates indicate the effect of economic instability on damage to community structure, which leads to displacement and crime. Liquor outlet density refers to the number of licensed liquor retailers per 12000 residents and measures environmental risk factors. Previous research shows the availability of alcohol is associated with violence due to impulsiveness. The location of CitiWatch surveillance cameras is included as a measure of police intervention and social surveillance in crime prevention, exploring whether increased surveillance in a particular neighbourhood is associated with reducing crime. Juvenile arrest data was also included to measure the average number of arrests per 1,000 people across communities. This variable reflects youth participation in criminal activities, which affects community stability and long-term crime rate. Moreover, neighbourhoods and years are considered predictors for interpreting local patterns and crime dynamics, which differ based on time and region.

This study introduced a web-scrapping dataset of more than 2,000 mental health service institutions in Baltimore. The data was collected from the 988 Suicide and Crisis Lifeline website and included the name, phone number, and address for every mental health service in Baltimore. The resulting dataset was geocoded and correlated with community-level statistics to explore the relationship between mental health service access, income levels, and violent crime rates.

2.2 Methods

This study applied the ordinary least squares (OLS) regression and machine learning method. The baseline linear relationships were established with OLS and statistical significance as testing using the p-value. Regression trees and random forest models are applied to capture the non-linear relationships, identify the interaction effects, and indicate the thresholds for selected variables. The random forest mode also shows an importance matrix to rank the importance of different predictors, providing different insights into the major factors influencing violent crime patterns. All the missing data and numerical variables are normalized. The interaction term in the regression table ensures the stability of the model using the cross-validation method. Regularization parameters in the regression tree models are calibrated to balance between minimizing overfitting and maintaining interpretability.

3. Summary Statistics

This section provides an overview of the dataset and visualization highlighting the socioeconomic and geographic disparities in violent crime in Baltimore from 2012 to

2017. The first subsection describes the primary variables through the summary table, and the second subsection reveals the relationship between the main socioeconomic factors and violent crime rate through scatter plots and maps. The third subsection describes the additional essential factors in crime activities of the web-scraping technique.

3.1 Summary Table

Variable	Count	Mean	Std. Dev.	Min	Max
Income	231,767	41,932.07	19,657.17	13,478.16	113,496.14
Dropout Rate	231,767	3.24	1.89	0.00	10.34
Violent Crime	231,767	20.45	13.94	0.54	97.58
Unemployment	231,767	14.47	6.48	1.99	29.91
Total Population	231,767	12,263.45	4,411.84	9,039	23,557
Mortgage Foreclosure	263,974	1.73	0.88	0.00	4.65
Liquor Outlet Density	231,767	1.48	1.62	0.00	8.53
Juvenile Arrest Rate	263,974	62.26	85.97	0.07	450.00

Table 1: Summary Statistics of Key Variables

The dataset includes more than 230000 observations collected at the community level. This table summarizes the eight key variables: income level, dropout rate, violent crime, unemployment rate, total population, mortgage foreclosure rate, liquor outlet density and juvenile arrest rate. The mean household income in Baltimore is $41932, ranging from $13478 to $113496, indicating a wide disparity with the high standard deviation of $19657. The mean dropout rate is 3.24%, and 10.34% is the maximum, suggesting that education is challenged in some areas. The mean violent crime rate per 1000 residents is 20.45 incidents and peaked at 97.58, indicating a high concentration of crime in specific high-risk neighbourhoods. There was a similar trend in the unemployment rate, with a mean value of 20.45%, which mentions a possible link between unemployment and the violent crime rate. The mean mortgage foreclosure rate is 1.73, ranging from 0 to 4.65, indicating a change in housing instability in Baltimore neighbourhoods. The mean liquor outlet density is 1.48 per 12,000 residents and peaked at 8.53, highlighting the obvious difference in alcohol supply across neighbourhoods. The juvenile arrest rate showed a high standard deviation of 85.97, with a maximum of 450 per 1000, reflecting extreme disparities in juvenile criminal activity. This table shows wide economic differences between different neighbourhoods, suggesting their impact on crime patterns. The web-scraped mental health data are excluded due to format incompatibility but will be analyzed in later sections.

3.2 Visualization

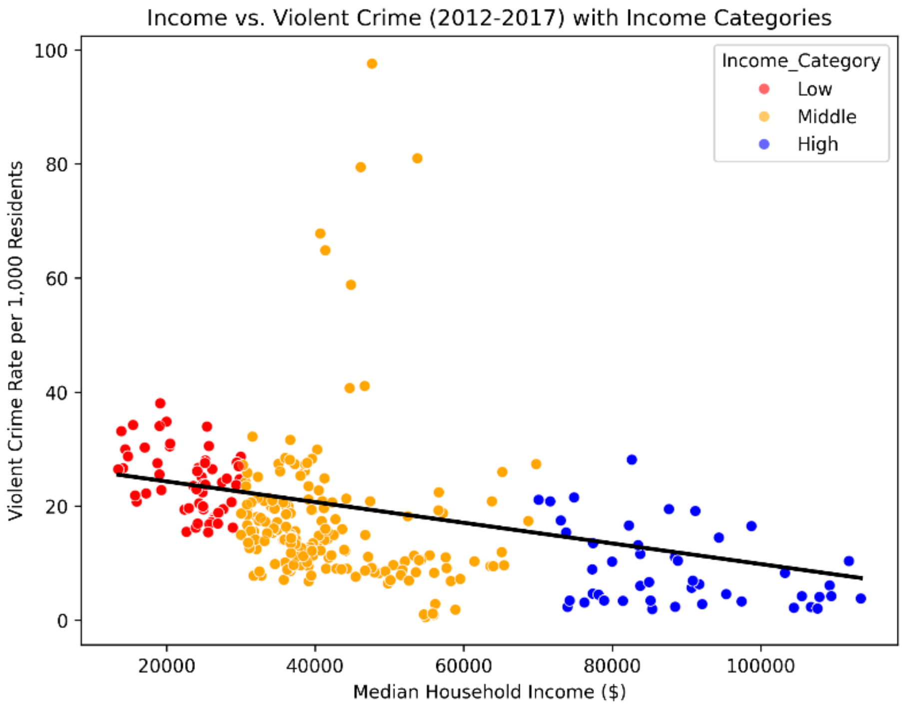

Figure 1: Income vs. Crime

Figure 1 shows that low-income neighbourhoods have significantly higher violent crime rates, while high-income neighbourhoods typically have crime rates below 20, a difference of nearly threefold. The negative relationship between income and violent crime rate almost remains constant, but the extreme outliers in middle-income areas suggest additional impacting factors.

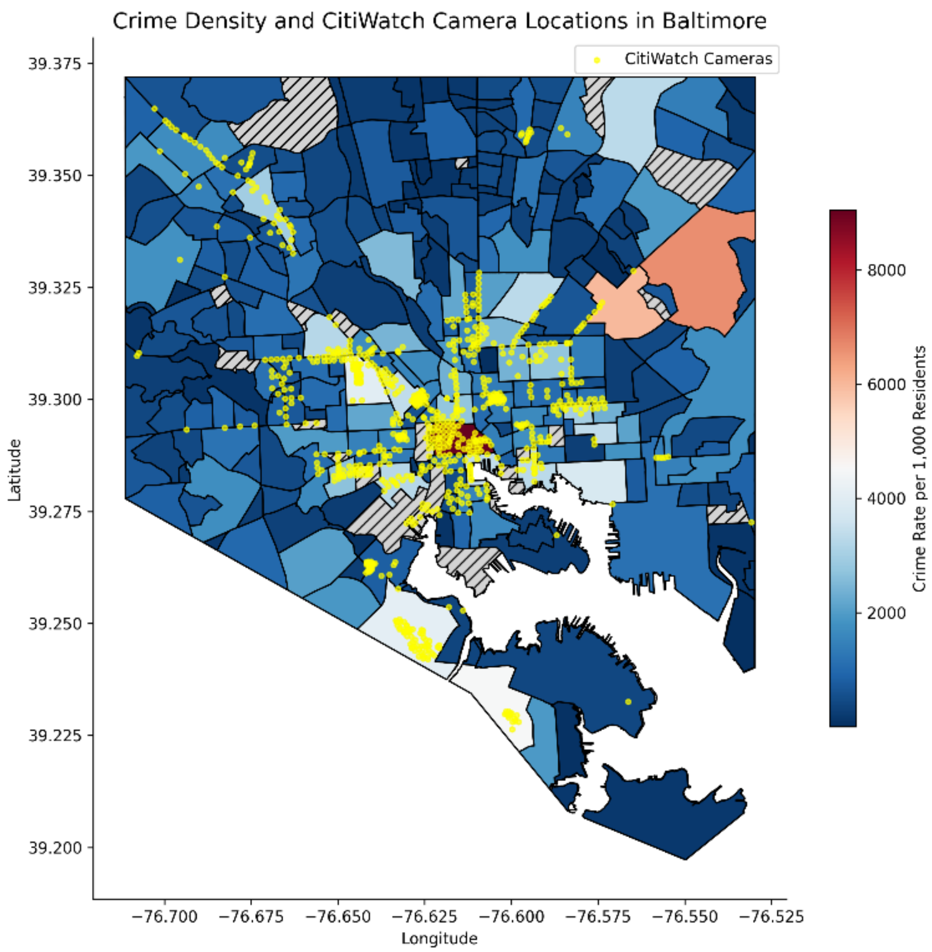

Figure 2: Crime Density & CitiWatch Cameras

Figure 2 shows the distribution of the violent crime density per 1000 residents and overlays the location of Citiwatch cameras throughout Baltimore. The high-crime areas, especially the downtown area, have crime rates of more than 8000 per 1000 residuals, while low-crime areas have crime rates of less than 2000. Surveillance cameras are concentrated in high-crime areas, but gaps remain in some high-crime-rate neighbourhoods. This aligns with the previous finding that higher crime in low-income areas is due to limited economic conditions. The relationship between crime density and surveillance distribution shows the target policing strategy but also raises the question: Is the high crime rate caused/deterred by cameras, or are they placed in areas already exercising high crime rates?

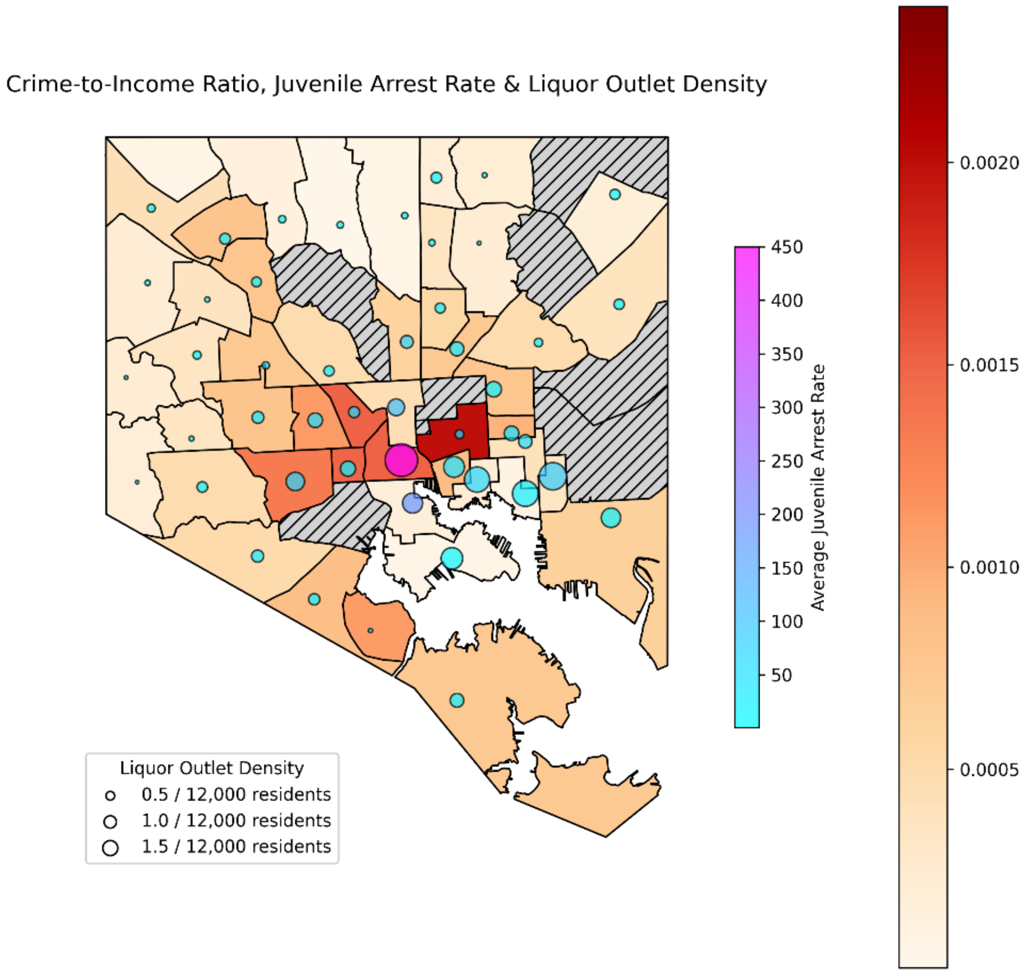

Figure 3: Crime-to-Income Ratio and Socioeconomic Risk Factors

Figure 3 shows the Crime-to-Income Ratio in Baltimore neighbourhoods to explore additional socioeconomic risk factors, including the juvenile arrest rate and the Liquor outlet density. Neighbourhoods with higher CIR tended to overlap with high juvenile arrest rates and liquor outlet density, which reinforces that economic distress and social disruption will lead to violent crime. In some center areas, the concentration of juvenile arrests is extremely high, up to 450 per 1000 juveniles, alongside a high density of liquor outlets (more than 1.5 per 12000 residents), driving the crime condition. The map highlights the relationship between economic instability and crime, consistent with economic theory of social disorder and criminal environment. However, some neighbourhoods with less social risk still show higher CIR, suggesting another contributing factor, such as the effect of geographical location or population density.

3.3 Web-Scraping Visualizations

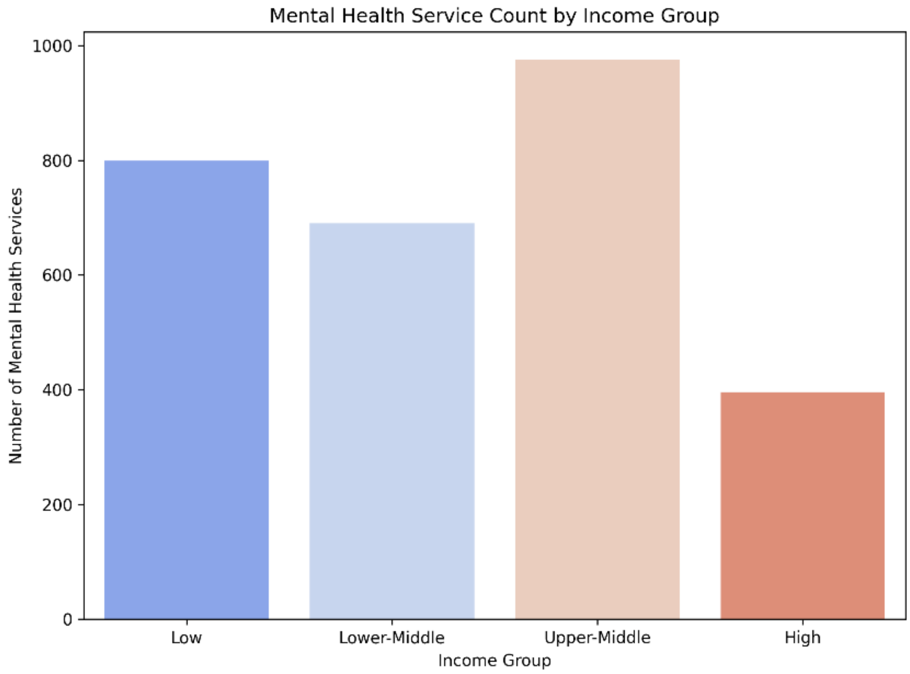

Figure 4: Mental Health Service Density by Different Income Group

Figure 4 shows the total number of mental health service locations across Baltimore neighbourhoods grouped by four different income levels. The upper-middle-income neighbourhoods have the highest number of mental health services, with nearly 1000, followed by low-income neighbourhoods, with about 800 and lower-middle-income neighbourhoods, with about 700. In contrast, the high-income communities have the fewest services, with fewer than 400. This distribution suggests that mental health services are more distributed in low and middle-income neighbourhoods, where the need is more significant due to the higher violent crime rate. There is a large gap between the upper-middle- and high-income groups with more than 580. From an economic perspective, the obvious difference in the accessibility of mental health services between different income groups raises problems about whether distribution is proportional to demand. This plot links to the research question by highlighting the uneven distribution of social infrastructure, a key component when considering how income inequality interacts with crime and vulnerability at the community level.

4. Results

4.1 OLS Regression Tables

The regression tables provide a detailed understanding of how economic and social factors conspired to influence violent crime rates in Baltimore from 2012 to 2017. Two main regression tables were estimated: one contains the main socio-economic determinants, and the other incorporates interaction terms and policy-relevant indicators.

Dependent variable:	ViolentCrime			
	(1)	(2)	(3)	(4)
Dropout			59.261***	-99.185***
			(15.387)	(12.601)
Income	-0.022***	-0.040***	-0.042***	-0.047***
	(0.001)	(0.002)	(0.002)	(0.002)
LiquorOutletdensity				788.107***
				(17.471)
MortgageForeclosure			-198.829***	-147.195***
			(36.569)	(28.785)
Unemployment		-82.455***	-80.996***	-26.353***
		(6.934)	(6.900)	(5.561)
const	2441.231***	4384.895***	4619.229***	3466.799***
	(68.425)	(176.651)	(203.670)	(162.215)
Observations	3304	3304	3304	3304
R^2	0.075	0.113	0.130	0.462
Adjusted R^2	0.074	0.112	0.129	0.461
Residual Std. Error	1709.761 (df = 3302)	1674.528 (df = 3301)	1658.931 (df = 3299)	1304.783 (df = 3298)
F Statistic	266.784*** (df = 1; 3302)	209.771*** (df = 2; 3301)	122.958*** (df = 4; 3299)	565.988*** (df = 5; 3298)

Note: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

Table 2: Regression Result - Violent Crime vs Socioeconomic Factors

The first regression table shows a significant negative relationship between income and violent crime, with each $1000 increase in income reducing violent crime by 4.7 units per community (Model 4). Aggregating the crime data by the community on a monthly basis reduces individual-level noise. The results show that liquor outlet density had a significantly positive association (788.1), enhancing the effect of social environment factors on the crime rate. Interestingly, the mortgage foreclosure rate shows a negative relationship to the violent crime rate, possibly due to lower residential density, which reduces potential victims. The inclusion of additional variables improves the model's explanatory power, increasing adjusted R² from 0.074 for Model 1 to 0.461 for Model 4, indicating that the socioeconomic and environmental factors together can better explain the crime pattern.

Dependent variable:	ViolentCrime			
	Model 0	Model 1	Model 2	Model 3
const	51.791***	33.372***	22.994***	13.872***
	(0.284)	(0.198)	(0.167)	(0.196)
Income_10k	-4.220***	-4.377***	-2.822***	-1.374***
	(0.036)	(0.024)	(0.019)	(0.028)
Unemployment	-1.074***	-0.329***	0.035***	0.430***
	(0.010)	(0.007)	(0.005)	(0.007)
Dropout	0.739***	0.070***	0.494***	0.736***
	(0.025)	(0.017)	(0.013)	(0.013)
LiquorOutletdensity		6.781***	0.883***	0.302***
		(0.022)	(0.027)	(0.027)
JuvenileArrestRate			0.123***	0.250***
			(0.000)	(0.002)
MortgageForeclosure			-1.129***	-1.849***
			(0.028)	(0.028)
Income10k_Juvenile				-0.017***
				(0.000)
Unemp_Juvenile				-0.006***
				(0.000)
Observations	77280	77280	77280	77280
R^2	0.168	0.632	0.809	0.826
Adjusted R^2	0.168	0.632	0.809	0.826
Residual Std. Error	12.643 (df=77276)	8.406 (df=77275)	6.062 (df=77273)	5.775 (df=77271)
F Statistic	5206.906*** (df=3; 77276)	33219.593*** (df=4; 77275)	54460.214*** (df=6; 77273)	45999.624*** (df=8; 77

Note: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

Table 3: Regression Result - Crime Policy, Social Environment, and Violent Crime

Table 3 shows the impact of economic and social environmental factors on violent crime. Income is scaled in $10000 units to improve interpretability and remains strongly negative, with a 10000 dollar increase in median income decreasing violent crime by 1.374 (Model 3). The coefficient of the unemployment rate becomes positively significant when additional variables are added, indicating that economic instability increases the risk of crime when other factors are considered. Dropout rates and juvenile arrest rates are increasing significantly, highlighting the importance of youth-related risks. Liquor outlet density remains positive, supporting the effect of social disorder. The interaction term between unemployment and juvenile arrest rate is negative (-0.006), indicating their combined effect may moderate individual impacts. These results support the hypothesis that violent crime stems not only from direct economic hardship but also from complex social factors.

4.2 Regression Trees

Regression trees analysis complements OLS regression by revealing nonlinear interaction and conditional relationship that would be difficult to observed.

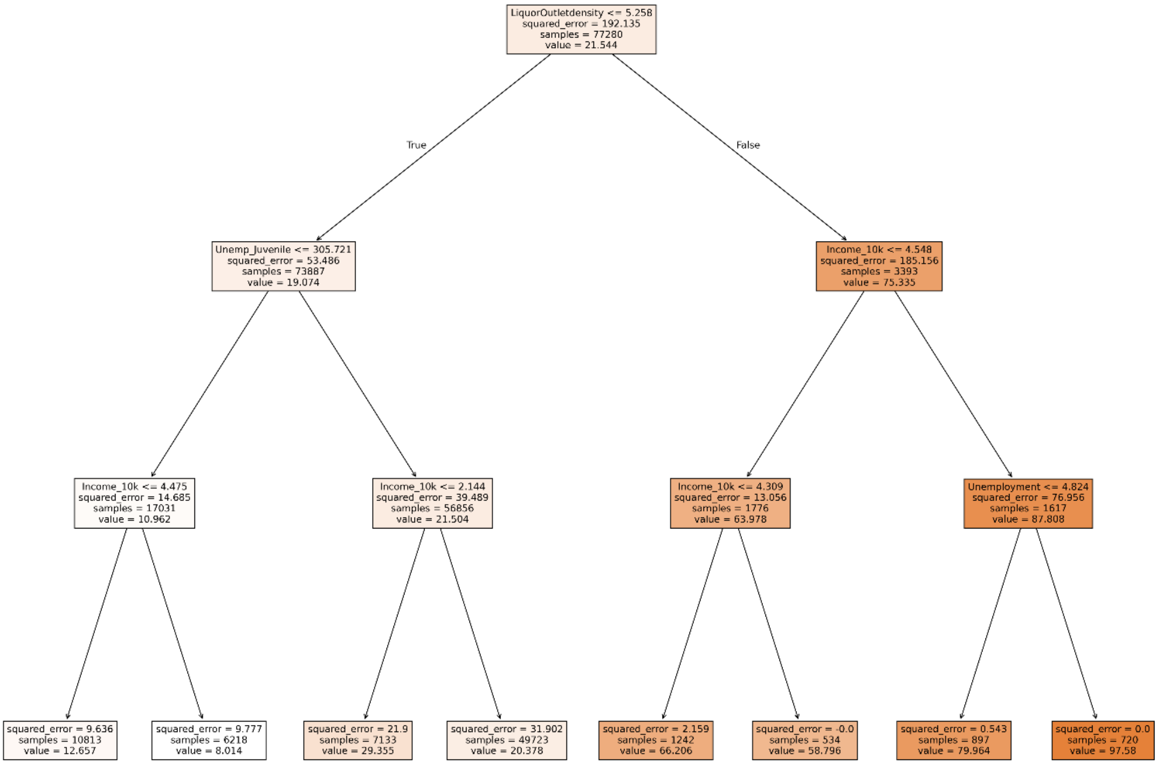

Figure 5: Regression tree on preferred OLS regression

The first regression tree used my preferred OLS regression variables, focusing on income (in $10000), unemployment, the Liquor Outlet Density, and the interaction between Unemployment and Juvenile Arrest Rate. The tree selects variables and thresholds that minimize the residual sum of squares (RSS) at each split in predicting violent crime across neighbourhoods. Liquor Outlet Density is the most influential predictor. Neighbours with less than 5.26 outlets per 12000 residents are on the left, with a lower average violent crime rate of 19.07 per 1000 residents, showing the strong impact of this factor. In the low-density branch, neighbourhoods with juvenile-related unemployment below -305.72 had lower crime rates, and the final income split shows a further decline in violent crime in neighbourhoods with an average income of above $44750. This split-sequence illustrates that the economic stress of youth is manageable when alcohol levels are low, and violent crime is reduced to a minimum when incomes are relatively high. Conversely, neighbourhoods with average incomes below $45,480 and unemployment rates above 4.8% had the highest crime rate of 97.58 per 1000 residents, highlighting the compounding effects of economic and social risk factors.

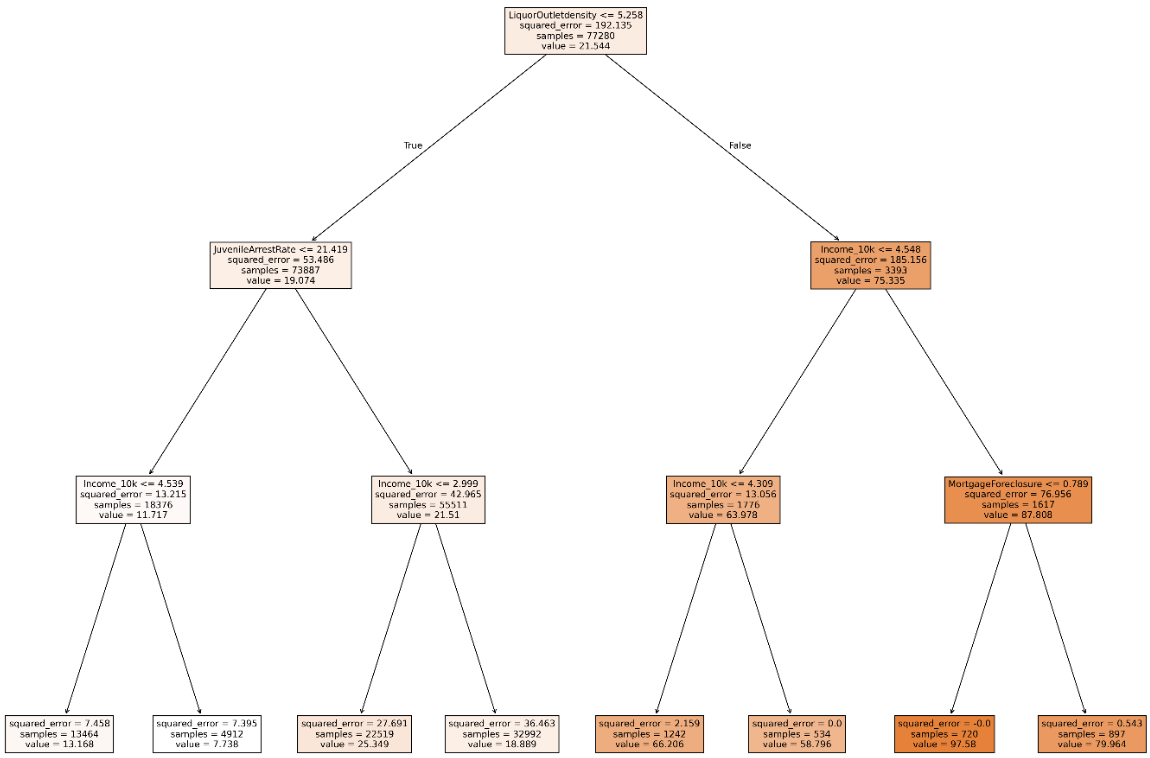

Figure 6: Regression tree on all numerical x variables

The second regression tree uses all numerical x variables, reiterating the importance of liquor export density, but the juvenile arrest rate is introduced as a second split, and the areas with an arrest rate of less than 21.42 per 1000 juveniles have a lower crime rate. Income became the third key factor, and areas with income above $45390 had the lowest crime rate of 13.17, and areas below this threshold showed a sharp increase in predicted crime rate. On the side of higher liquor outlet density, mortgage foreclosures are the third key factor. The highest crime rate is nearly 98 in neighbourhoods with a low average income (below $43090) and high alcohol density. The strength of a tree model is its ability to indicate obvious thresholds and capture interaction effects between predictors without modelling them explicitly. It reveals nested factors that may be difficult to identify with simple OLS.

4.3 Random Forest

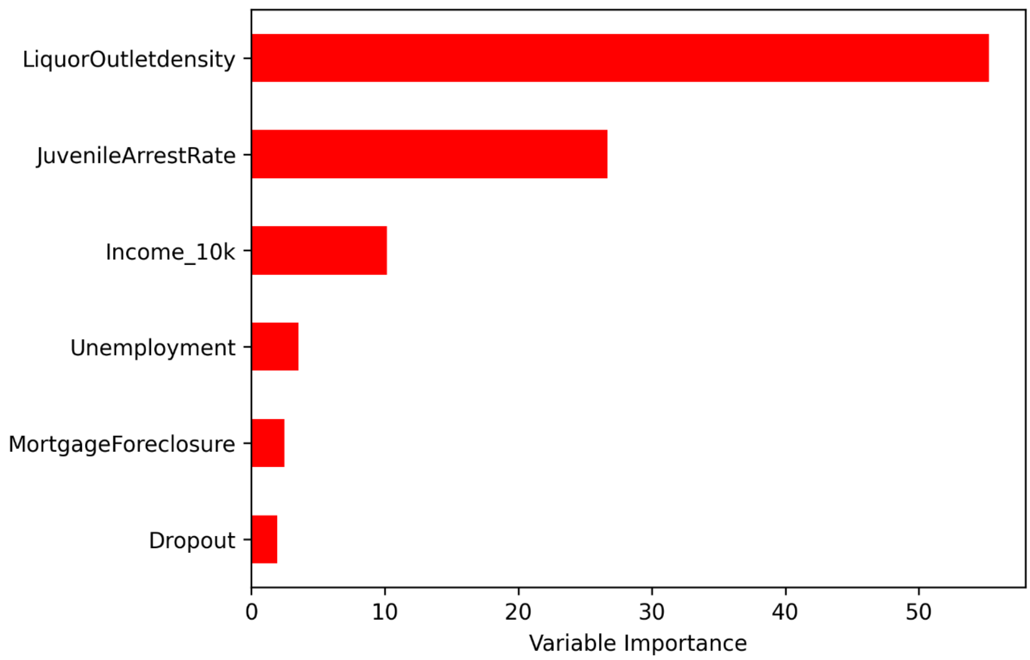

Figure 7: Importance Matrix

The Random Forest Importance Matrix shows that liquor outlet density and juvenile arrest rates are the two most influential predictors of violent crime, income level (main predictor) ranks only third to liquor outlet density and juvenile arrest rates. This result is a little surprising since economic theoretical prediction emphasizes income in explaining violent crime. The lower ranking of income may reflect the interaction and non-linear effect, indicating that access to liquor and juvenile activity have a more direct impact on violent crime across neighbourhoods. Income still plays a role, but social factors may partly mediate its effect. For example, low-income neighbourhoods tend to have more liquor outlets and higher rates of juvenile arrest. This suggestion that income has a more indirect effect on crime highlights how income inequality can be a significant driver through social factors that lead to violent crime. The random forest helps aggregate multiple trees, reducing variance and overfitting risk. Combining the results from the regression trees, the finding emphasizes that while income is critical, policy interventions may need to prioritize and consider social and environmental risks. Violent crime is driven by a combination of economic and social factors, not just income levels.

5. Conclusion

This paper explores how income inequality impacted violent crime rates significantly in Baltimore from 2012 to 2017. While the relationship between poverty and crime is well-documented, this study highlights the importance of economic and social factors in explaining violent crime at the community level. The findings suggest that income plays a significant role, but this relationship is not entirely linear and regulated by social and environmental factors, consistent with previous studies (MacDonald, 2015; Heagerty, 2010).

Regression trees and random forest models are used in this research to solve this nonlinear problem. Regression trees are particularly effective for identifying thresholds, potential interaction effects, and nested relationships without specifying these interactions in advance. For example, the tree shows that violent crime rates rise sharply only when both alcohol export density and youth unemployment exceed certain thresholds. Random forest improves prediction accuracy by reducing overfitting and variables from multiple regression trees. In this study, the random forest model provides a ranking of the importance of predictors, revealing that liquor outlet density and juvenile arrest rates contribute more to predicting violent crime than income itself, indicating the indirect effect of income on violent crime. Additionally, the study introduces a new dataset that scraped mental health service information from the 988 Suicide & Crisis Lifeline website. Based on these data, crime and income visualizations show the services in high-crime, low-income areas support the theory that inequality in social infrastructure may exacerbate crime (Chalfin & McCrary, 2017).

To reduce the violent crime rate, policymakers must address its economic root issue. According to labour market theory, raising the minimum wage and expanding social welfare programs can reduce the financial pressure that stimulates crime. Restricting alcohol sales in high-crime areas can also reduce economic risk, consistent with rational choice theory, to defer criminal behaviour. These are not just statistics but families in need of support. Future research should refine this insight with more comprehensive data to help shape policies that can create long-term solutions to reduce crimes in the most affected communities.

References

1. Yildiz, R., Ocal, N., & Yildirim, J. (2013). The effects of unemployment, income, and education on crime: Evidence from individual data. *Journal of Economic Behavior & Organization, 89*, 135-152.

2. Fajnzylber, P., Lederman, D., & Loayza, N. (2002). Inequality and violent crime. *Journal of Law and Economics, 45*(1), 1-39.

3. Sampson, R. J., Raudenbush, S. W., & Earls, F. (1997). Neighborhoods and violent crime: A multilevel study of collective efficacy. *Science, 277*(5328), 918-924.

4. Grogger, J. (1998). Market wages and youth crime. *Journal of Labor Economics, 16*(4), 756-791.

5. Gillani, S. Y. M., Rehman, H. U., & Gill, A. R. (2009). Unemployment, poverty, inflation, and crime nexus: Cointegration and causality analysis of Pakistan. *Pakistan Economic and Social Review, 47*(1), 79-98.

6. Kelly, M. (2000). Inequality and Crime. *The Review of Economics and Statistics*, *82*(4), 530–539. http://www.jstor.org/stable/2646649

7. Raphael, S., & Winter-Ebmer, R. (2001). Identifying the Effect of Unemployment on Crime. *The Journal of Law & Economics*, *44*(1), 259–283. https://doi.org/10.1086/320275

8. Becker, G. S. (1968). Crime and punishment: An economic approach. *Journal of Political Economy, 76*(2), 169-217. https://doi.org/10.1086/259394

9. Lochner, L., & Moretti, E. (2004). The Effect of Education on Crime: Evidence from Prison Inmates, Arrests, and Self-Reports. *The American Economic Review*, *94*(1), 155–189. http://www.jstor.org/stable/3592774

10. Ehrlich, I. (1973). Participation in illegitimate activities: A theoretical and empirical investigation. *Journal of Political Economy, 81*(3), 521-565. https://doi.org/10.1086/260058

11. Grogger, J. (1998). Market Wages and Youth Crime. *Journal of Labor Economics*, *16*(4), 756–791. https://doi.org/10.1086/209905

12. American Economic Association. (2009). *American economic journal. Economic policy.* American Economic Association.

13. Fougére, D., Kramarz, F., & Pouget, J. (2009). YOUTH UNEMPLOYMENT AND CRIME IN FRANCE. *Journal of the European Economic Association*, *7*(5), 909–938. https://doi.org/10.1162/JEEA.2009.7.5.909

14. Lin, M.-J. (2008). Does Unemployment Increase Crime? Evidence from U.S. Data 1974-2000. *The Journal of Human Resources*, *43*(2), 413–436. http://www.jstor.org/stable/40057352

15. Heagerty, M. F. (2010). Crime and the Environment—Expanding the Boundaries of Environmental Justice. *Tulane Environmental Law Journal*, *23*(2), 517–535. http://www.jstor.org/stable/43294093

16. Shover, N., & Routhe, A. S. (2005). Environmental Crime. *Crime and Justice*, *32*, 321–371. http://www.jstor.org/stable/3488362

17. MacDonald, J. (2015). Community Design and Crime: The Impact of Housing and the Built Environment. *Crime and Justice*, *44*(1), 333–383. https://doi.org/10.1086/681558

Appendices

A Tables

Table 1. Regression Table - Violent Crime Income level and Unemployment

This regression table shows the relationship between income levels, unemployment and violent crime in Baltimore. The negative coefficients on IncomeMid and IncomeHigh in Model 1 support the economic theory that a higher income level has lower violent crime rates, consistent with Becker's (1968) economic model on criminal motivation. Model 3 shows that high-income neighbourhoods had an average of 32.158 fewer violent crimes per 1000 residents than low-income neighbourhoods (baseline). The positive interaction between unemployment and high income (0.652) indicates that unemployment increases violent crime, but the effect is less severe in wealthier areas. Conversely, the negative interaction between unemployment and income (-1.091) in Model 2 indicates unemployment substantially impacts crime in middle-income neighbourhoods. Model 2 has the more significant value of adjusted R^2, indicating stronger explanation power and additional interaction terms improve the model fit. Targeting unemployment in middle-income neighbourhoods may be most effective since sudden unemployment in previously stable lives creates greater feelings of anxiety, frustration, and injustice, potentially fueling crime.

Dependent variable:	ViolentCrime			
	Model 0	Model 1	Model 2	Model 3
const	21.834***	45.895***	30.419***	46.073***
	(0.071)	(0.128)	(0.213)	(0.129)
Unemployment	-0.095***	-1.019***	-0.268***	-1.028***
	(0.004)	(0.006)	(0.010)	(0.006)
IncomeMid		-11.170***	9.045***	-11.228***
		(0.072)	(0.235)	(0.073)
IncomeHigh		-28.600***	-17.070***	-32.158***
		(0.124)	(0.177)	(0.299)
Unemp_IncomeMid			-1.091***	
			(0.012)	
Unemp_IncomeHigh				0.652***
				(0.050)
Observations	231767	231767	231767	231767
R^2	0.002	0.188	0.215	0.188
Adjusted R^2	0.002	0.188	0.215	0.188
Residual Std. Error	13.922 (df=231765)	12.560 (df=231763)	12.346 (df=231762)	12.556 (df=231762)
F Statistic	456.79*** (df=1;231765)	17850.86*** (df=3;231763)	15888.83*** (df=4;231762)	13440.8*** (df=4;231762)

Note: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

B Figures

Figure 1. Crime rate per 1,000 residents by neighborhood
This bar chart shows the first ten neighbourhoods with the highest violent crime rate per 1000 residents using a colour gradient to distinguish crime intensity. Seton Hill (75.3) and Downtown (74.6) have the highest rates, compared to other areas at 32.5 and below. This highlights the concentration of crime in specific neighbourhoods, representing external factors, such as income level, unemployment, or education gap, that may contribute to the crime disparities across Baltimore.

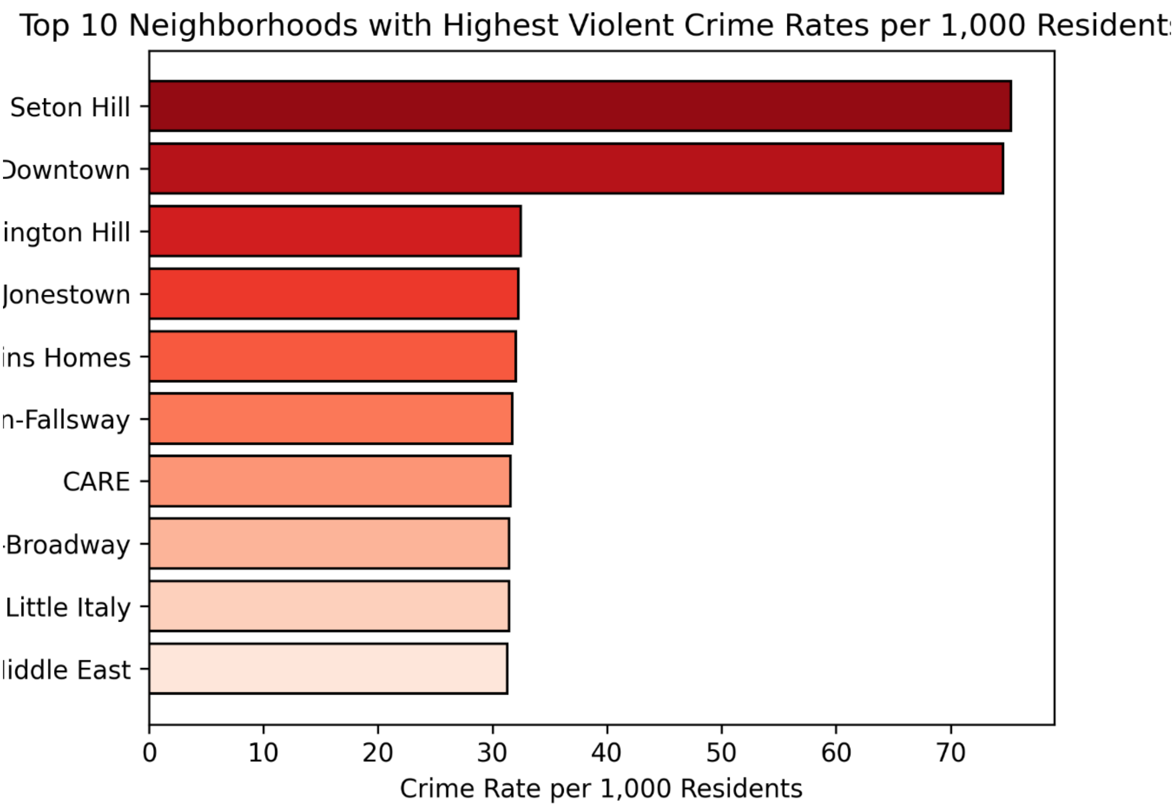

Figure 2. Violent crimes vs income (2012-2017)

The scatter plot examines the relationship between household income levels and violent crime rates from 2012-2017. A clear negative correlation indicated that low-income neighbourhoods tend to have higher violent crime rates. Neighbours with income below $40,000 have an average violent crime rate of 20-40 incidents per 1000 residents, while high-income areas with more than 80000 residents seldom have more than 20. The presence of outliers suggests that high crime rates are not only caused by low income, highlighting the necessity of exploring other confounder variables.

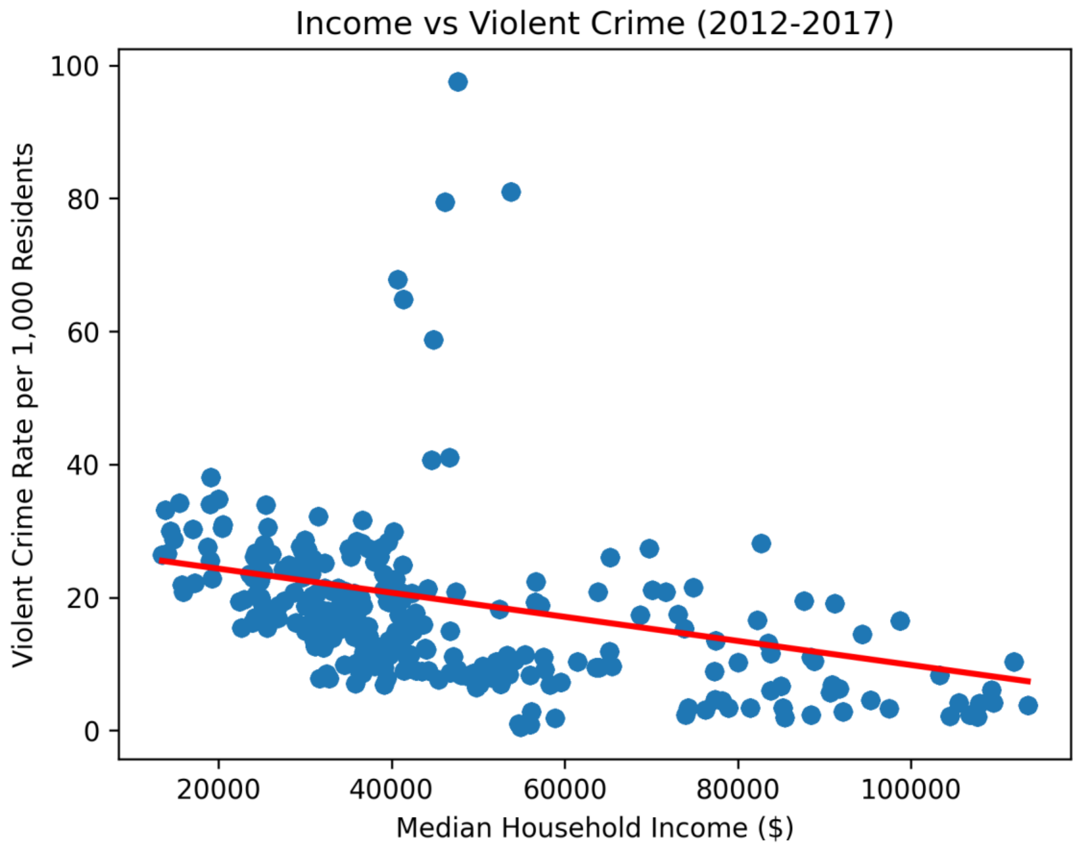

Figure 3. Income Distribution across the year

The violent plot visualizes the distribution of median household income in Baltimore from 2012 to 2017. The width indicates the higher density of income values, and the long tail shows income inequality(outlier). The median income fluctuates around $40,000, with the highest density between $30,000 and $500,000. The longest tail extends over $100,000 in 2017, indicating income inequality. Low-income families (under $25,000) have existed for years, highlighting economic disparities that may lead to higher crime rates in poor areas.

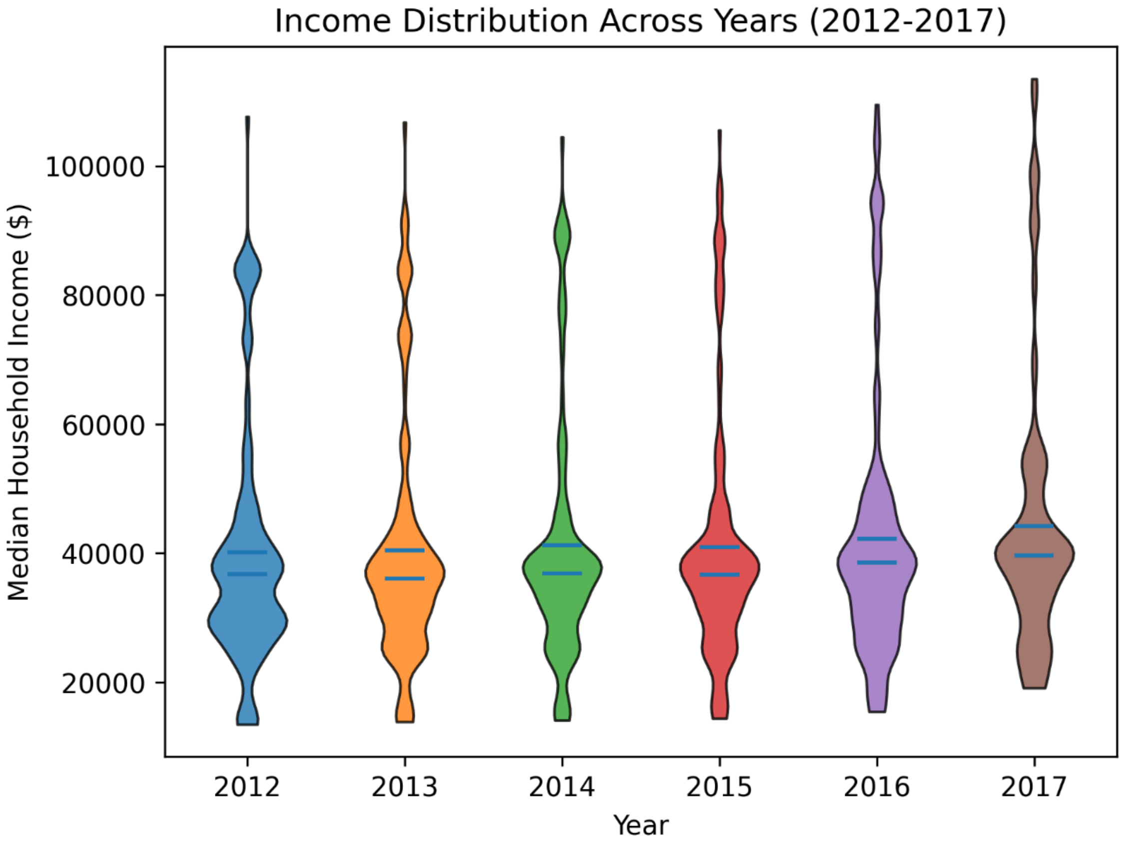

Figure 4. Trend of High School Dropout Rates (2012-2017)

The line graph shows the change in the high school dropout rate from 2012 to 2017, with a maximum value of 4.0 percent in 2012 and 2015 and a minimum of 2.2 percent in 2013 and 2014. Reports from the Maryland Department of Education and local news sources highlight this sharp decline from 2013 to 2014, especially among minority students. The fluctuation in dropout rates indicates instability in education levels, which may be linked to crime patterns. Higher dropout rates may indicate an economic decline, affecting higher crime rates in lower-income neighbourhoods.

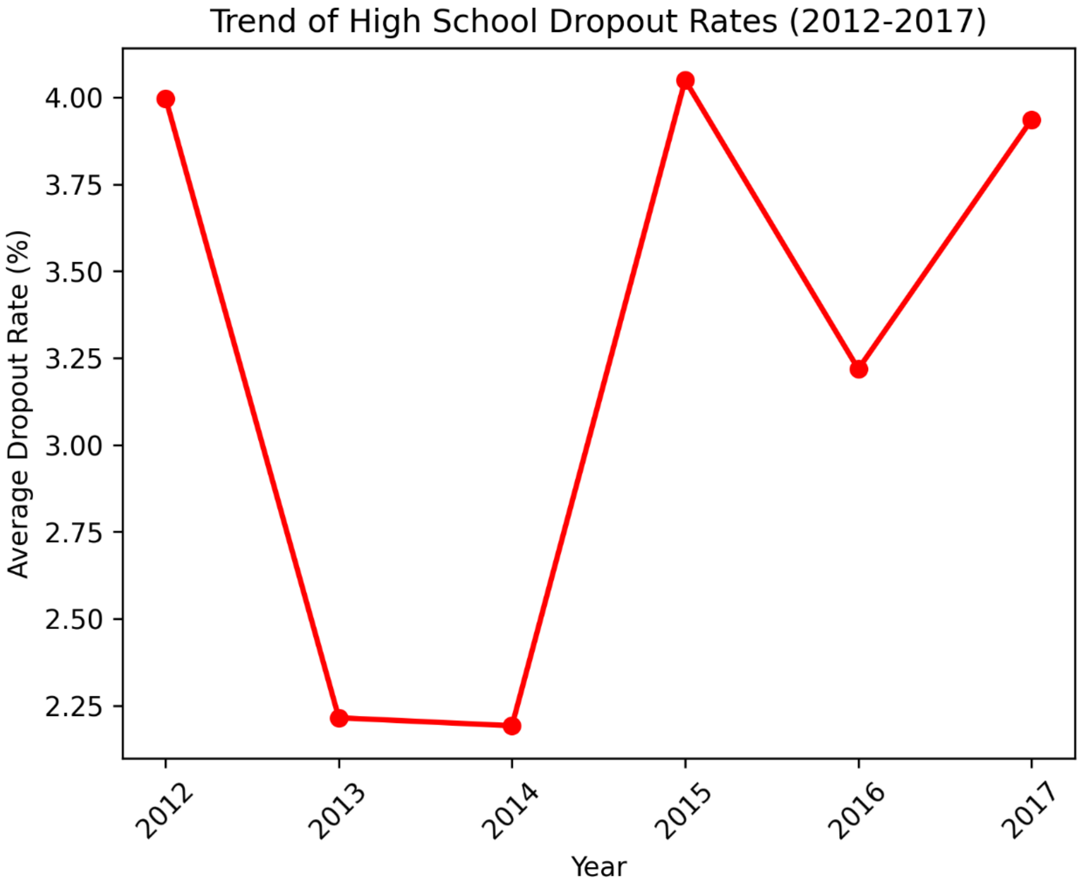

Figure 5. Comparing crime rates across income groups in 2017

This box plot compares violent crime rates across different income groups. Low-income areas have the highest median violent crime rate, around 28 per 1000 residents, while high-income neighbourhoods have a lower median, below 15 per 1000 residents. However, the wide distribution range, especially in high-income neighbourhoods(outliers above 80), highlights differences in the distribution of crime. This raises the question of whether these high-crime outliers link to specific downtown neighbourhoods, which requires further investigation.

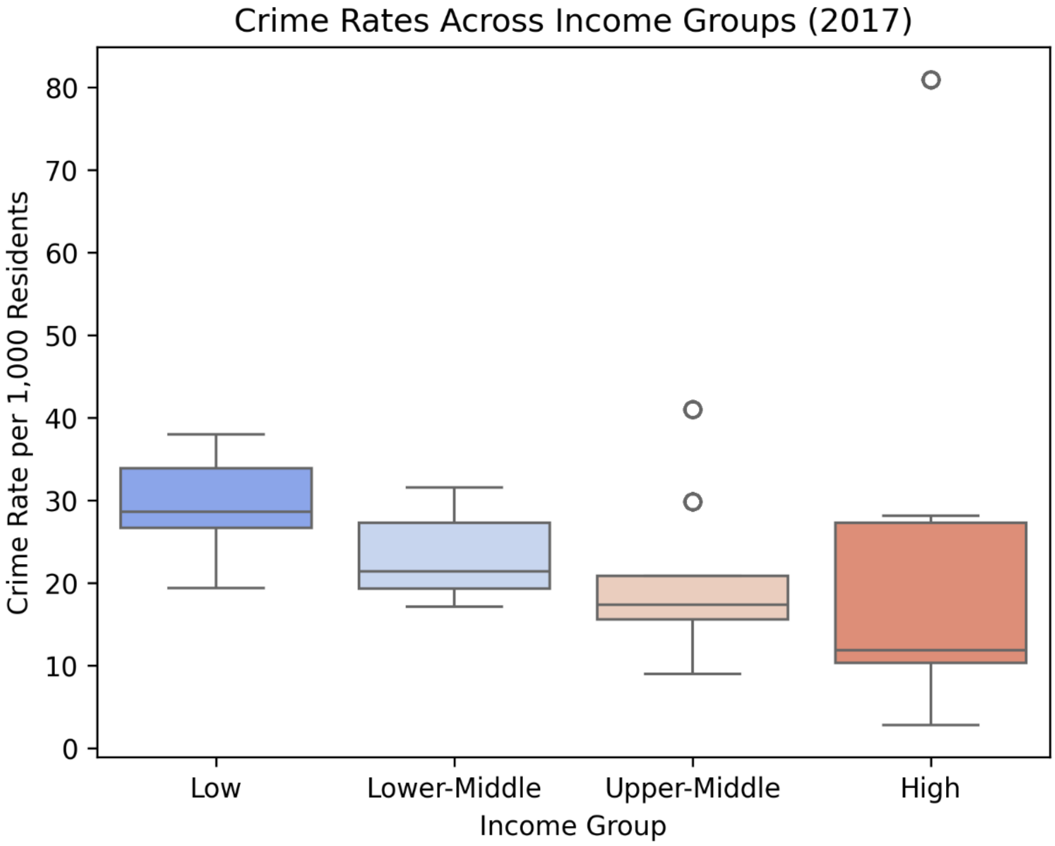

Figure 6. Side-by-Side Histograms for Violent Crime (2012-2017)

These histograms show the distribution of violent crime from 2012 to 2017, allowing the comparison of violent crime intensity across the time. Crime rates peak around 20 per 1000 residents, and some years show more significant variability, with extremes approaching 80. The fact that violent crime rates remain stable in the long term illustrates that systemic problems like income gaps sustain crime rates rather than short-term fluctuations in economic conditions.

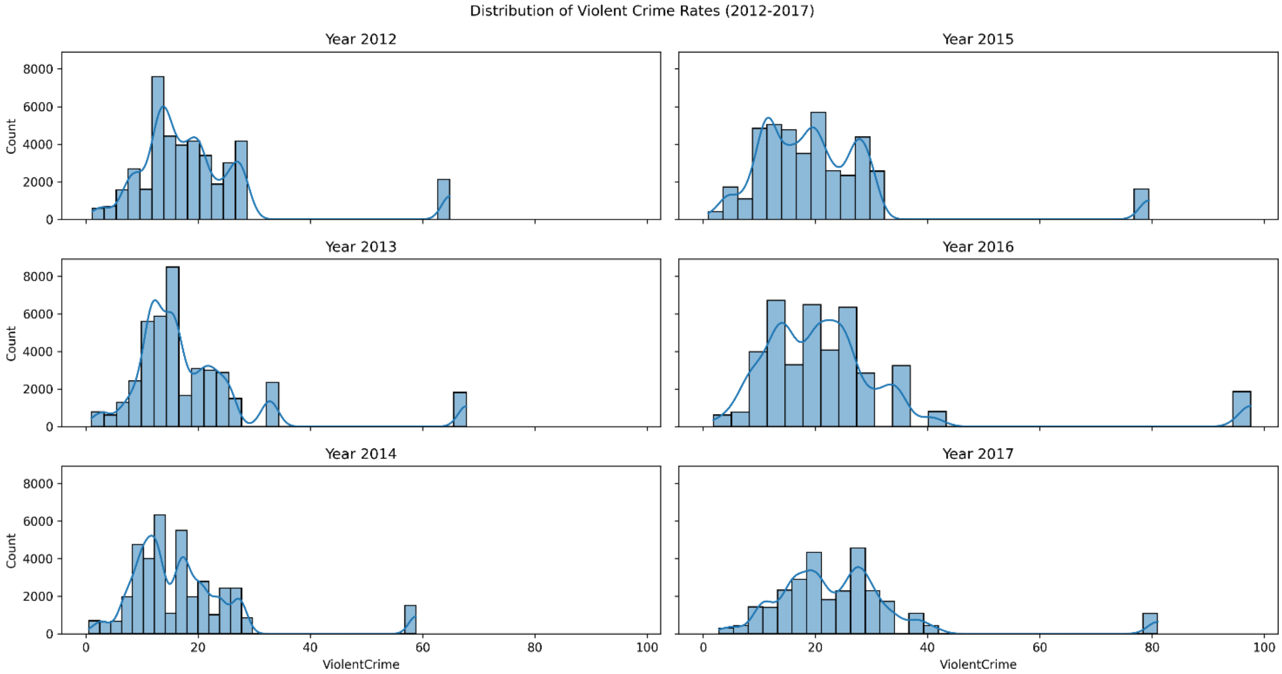

Figure 7. Relationship Between Unemployment & Violent Crime

This scatter plot shows the relationship between unemployment and violent crime from 2012 to 2017. There is a strong positive relationship between these two, with crime sharply rising as unemployment rises, reaching nearly 100 per 1000 residents at maximum. This consistent trend over the years shows that unemployment is also a significant factor in violent crime.

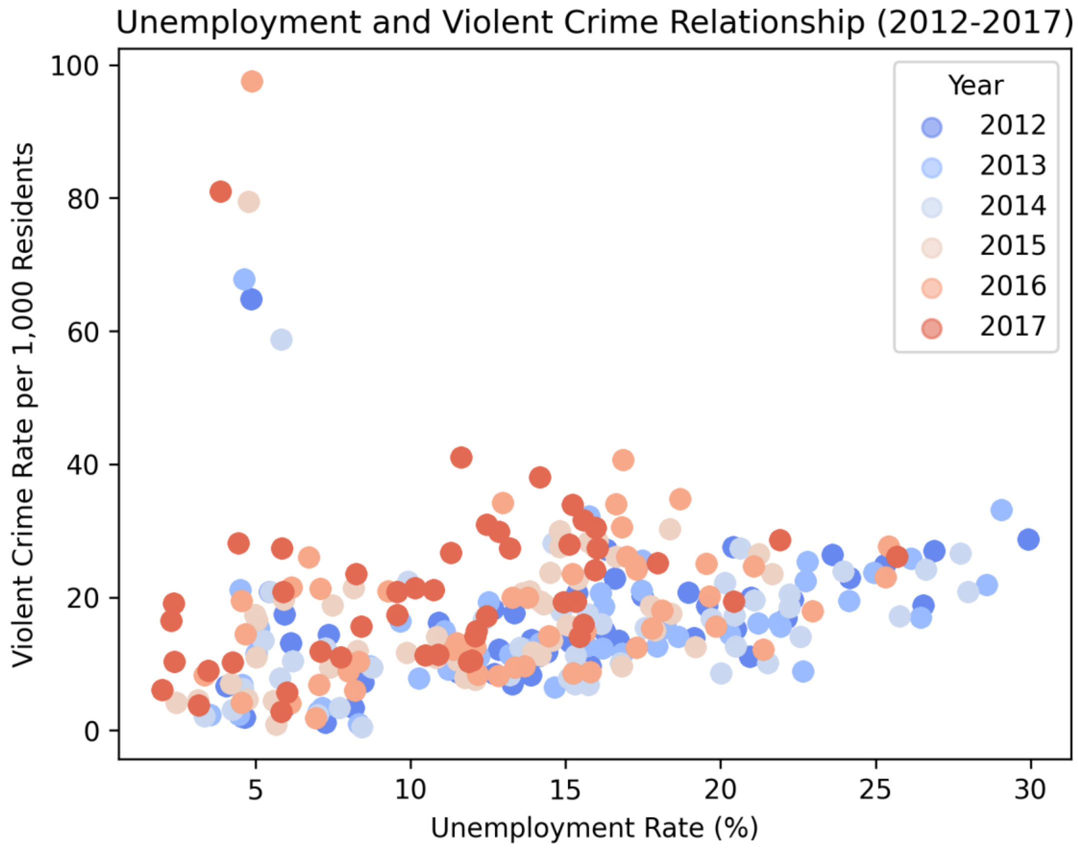

Figure 8. Map 1 - Violent Crime rate in Baltimore (Y)

This map shows the violent crime rate per capita in Baltimore, the dependent variable(Y) in the research question, highlighting areas with a high crime rate. Downtown has the highest crime rate, with more than 8 crimes per resident, and there are some unexpectedly high crime rates in northeastern communities. The map is based on community-level data and has some missing values. This map highlights the disproportionate crime rate in specific communities, reinforcing the link between crime burden and population density.

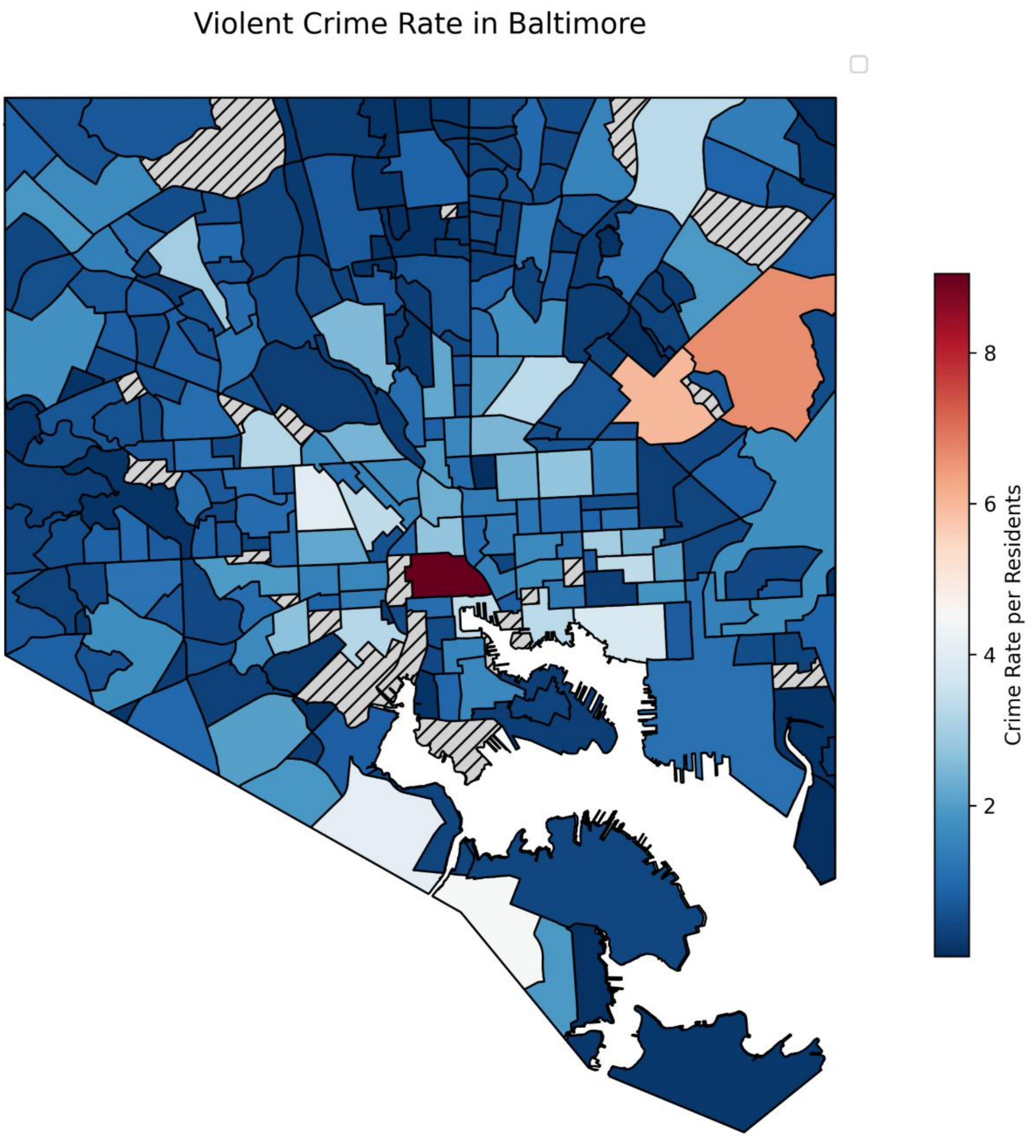

Figure 9. Map 2 - Income Level in Baltimore (main X)

This heat map shows the median household income earned in one year by neighbourhood in Baltimore. Dark red regions indicate the neighbourhoods with the highest income level (over 10000 dollars), and the light-yellow regions show the low-income neighbourhoods (under 20000 dollars). This dataset is based on the community level without obvious missing values. The income gap is clear, showing that Baltimore's northwest and downtown areas have higher income levels and lower income levels in the south and southeast regions. Combined with Map 1, the pattern is almost consistent with the economic theory that lower-income neighbourhoods have higher crime rates, reinforcing the impact of economic disparity on crime distribution. However, some mid-level income areas also have unexpectedly high violent crime rates, which may be investigated later.